PracTools Project

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Assigned 2/23/2026; Due 4/20/2026

Introduction

In this project, you will design a multi-stage sample of census tracts, block groups (BGs), and persons from the City of Detroit in the State of Michigan in the U.S. The sample will support a face-to-face survey that aims to examine transportation security and transportation use among Detroit residents. Key variables include the proportion of residents living in a household that does not own or lease vehicle (e.g., automobiles, vans, trucks) for use by household members, the proportion of residents whose main means of transportation is public transportation, the proportion of residents who skip going somewhere because of a problem with transportation, and the proportion of residents who asked for transportation help from friends, family or neighbors in the past month.

Based on considering analytic subgroups associated with age, the desired precision of estimates, and the available budget, it has been determined that the following sample sizes of respondents are to be selected:

Age Group	n (respondents)	Expected Response Rate
18-34 yrs	250	0.35
35-49 yrs	250	0.45
50-64 yrs	250	0.50
65+ yrs	250	0.65
Total	1000	
Sample tracts	40	
Sample blockgroups/tract	1	

The sample design will use census tracts as PSUs, block groups (BGs) as SSUs, and persons as elements. The goal of the sample design is to select a sample that will yield the responding sizes specified in the table above while 1) achieving a self-weighting sample in each of the age groups above and 2) obtaining an equal workload in each sample PSU.

You should pay particular attention to tracts with less than 2 BGs and to areas with small population counts and decide how they should be handled in the frame.

Use systematic sampling with probabilities proportional to size to select the PSUs and SSUs. Sort the file in order by tract before selecting the sample. This method of selection is available in the R `sampling` package. When using R to select the sample, include the statement

```
set.seed(-77)
```

at the beginning of your code.

Deliverables

Deliverables are required for the project:

1. A sampling report (details of the report below); file name: `REPORT_GROUPNAME.pdf`
2. Text files giving the units used for the area frames and relevant census counts and measures of size; file name: `FRAME#_GROUPNAME.pdf`
3. Text file for the selected sample along with relevant census counts, measures of size, selection probabilities, and weights; file name: `SAMPLE_GROUPNAME.pdf`

Sampling Report

Contents

This will be a **technical report** written to a client whose staff includes managers and technical personnel. Managers will be more interested in understanding the broad outline of the steps used in weighting. Technical personnel will be interested in understanding the details of sample selection and weight computation, including appropriate formulae. You should consider how to structure your report to serve these audiences.

Below is a list of topic areas that should be included in your report. The order of the sections in your report does not have to be the same as that given below. The report should be constructed in a way that presents topics in an order that you find logical. However, the report should include all topic areas that can be easily identified by the readers.

Topic Areas for the Sampling Report

1. Title Page (project title, date of submission, team name, names of team members, and team contact person's name and contact information)
2. Introduction (overview of the report)
3. Sample Design
   - Goals of the sample design
   - Target population
   - Area sampling frame: units, data available, source of the data
   - Assigning measures of size to units
   - Descriptive statistics on the frame you created
   - Discuss whether a design that uses the same sampling rate for all persons could be used to achieve the sample size targets for each age group. Describe any advantages or disadvantages that this design might have. Support your conclusions with numerical calculations.
4. Sample Selection
   - Method of selection
   - Selected units and characteristics of each
   - Selection probabilities of units at each stage of selection
   - Description of how persons should be selected from area listings
5. Anticipated Precision

    - Precision of estimated proportions of persons in Deroit who answer public transportation as the main means of transportation
6. Description of weighting plan
7. General explanation of how an analyst should estimate variances from the sample
    - Discuss how to define strata, PSUs, and any other characteristics that should be considered
8. Maps
    - The City of Detroit
    - Selected tracts and block groups
9. Appendix
    - Codebook of frame and sample files, i.e. a list of the variables on the files and a description of each variable
    - Listing of the sample PSUs and sample SSUs with their selection probabilities and census data. On each sample SSU, list the sampling rate you will use to select persons in each domain.
    - Disclosure of AI Use, including the purpose, AI models and prompts used (verbatim)

Format

- 10 to 15 pages, excluding the title page and the appendices
- Line spacing: Single
- 1 inch margins
- Font size 11 or larger
- Pages should be numbered
- Use informative section titles

Data files and other information

Similar report from the Detroit Metro Area Communities Study

Available at https://detroitsurvey.umich.edu/wp-content/uploads/2025/02/Transportation-Insecurity-in-the-Motor-City_FINAL.pdf

Background information about Census geographies

- Census geography hierarchy: https://www2.census.gov/geo/pdfs/reference/geodiagram.pdf
- Census geography definition: https://www.census.gov/programs-surveys/geography/about/glossary.html
- Collection of census geography reference maps: https://www.census.gov/programs-surveys/geography/geographies/reference-maps.html

Detroit, MI population data

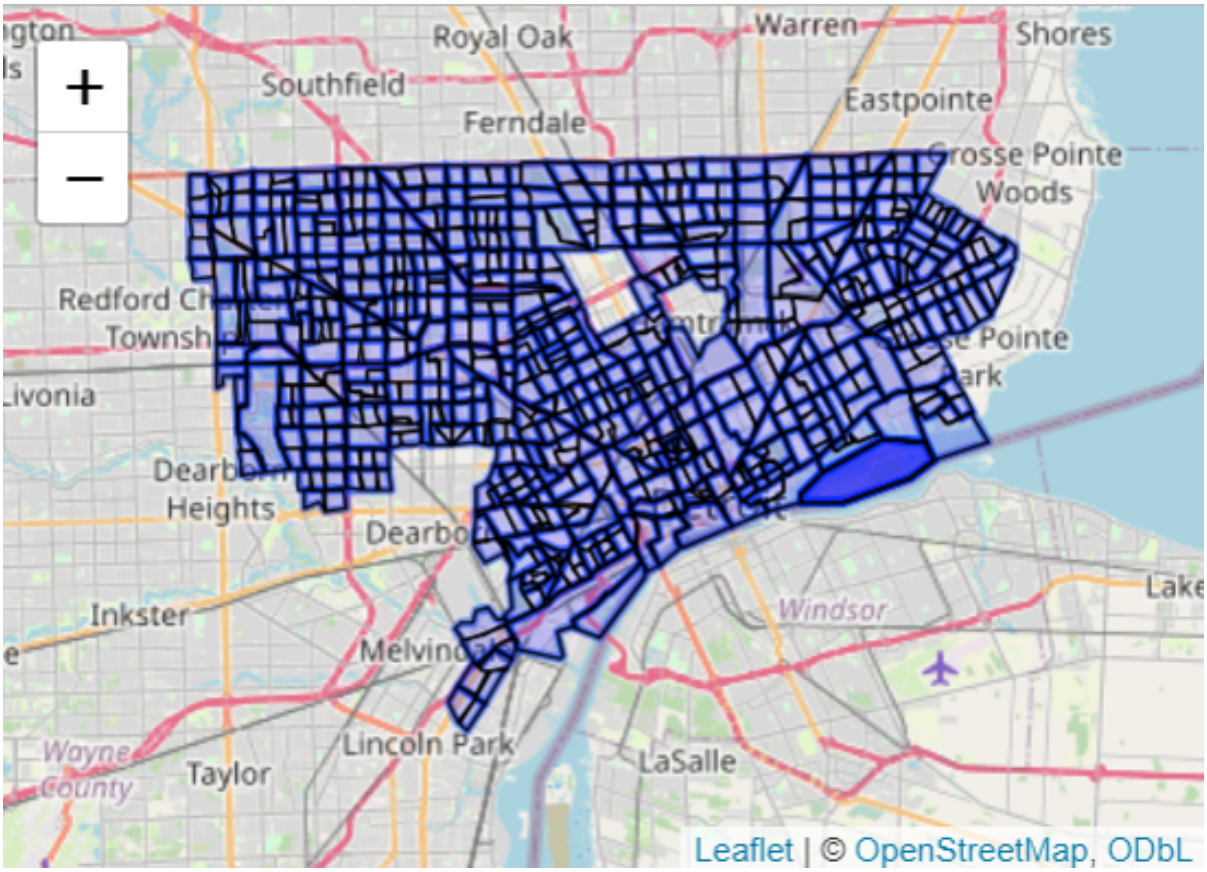
- Detroit MI.xlsx available on Canvas
  - Includes both Census tract- and blockgroup-level aggregate data on 1) housing counts, 2) person counts and 3) person counts by Sex x Age from the 2020 Decennial Census Demographic and Housing Characteristics (DHC) Summary File (SF). (Note that there are many types of SFs from 2020 Decennial Census; See https://api.census.gov/data.html for detail.)
  - Prepared by using `tidycensus` package to download the Census tract level SFs for Wayne County, MI (which include Detroit) and merging it with FIPS codes of Census tracts in Detroit in the code below.
  - Note: In order to download SFs using `tidycensus`, a census Application Programming Interface (API) key is required. You can obtain this from https://api.census.gov/data/key_signup.html.

```
library(tidycensus)
library(tidyverse)

# "Census Tract FIPS Code Detroit MI.csv":
# List of Census Tracts in Detroit obtained from
# City of Detroit Open Data Portal
# https://data.detroitmi.gov/

TractFIPS<-read.csv("YOUR DIRECTORY/Census Tract FIPS Code Detroit MI.csv",
                    header=T,
                    col.names=c("GEOID"),
                    colClasses = "character")
```

```r
# Download Decennial Census SFs

# Once you have the api key, you can set with:
census_api_key("YOUR KEY", install = TRUE)
# No need to set repeatedly.

# You can look up what variables are available in 2020 Decennial DHC
var2020dhc <- load_variables(2020, "dhc", cache = TRUE)
View(var2020dhc)
# From this, you can determine what variable you want.
# In our case, 1) housing counts (H1_001N),
#              2) person counts (many options) and
#              3) person counts by Sex x Age ("P_12" series).

# For tract level housing counts:
hh_tract<- get_decennial(geography ="tract",
                   variable=c(TotHH="H1_001N"),
                   year = 2020,
                   sumfile="dhc",
                   state="MI",
                   county="Wayne",
                   output="wide")%>%
  right_join(TractFIPS,by="GEOID")

# For tract level person counts -- Sex x Age:
person_tract<- get_decennial(geography = "tract",
                       table="P12",
                       year = 2020,
                       sumfile="dhc",
                       state="MI",
                   county="Wayne",
                   output="wide")%>%
  right_join(TractFIPS,by="GEOID")%>%
  rename(TotPerson    =P12_001N,
         TotMale      =P12_002N,
         MaleLT5Yrs   =P12_003N,
         Male5to9Yrs  =P12_004N,
         Male10to14Yrs=P12_005N,
         Male15to17Yrs=P12_006N,
         Male18to19Yrs=P12_007N,
         Male20Yrs    =P12_008N,
         Male21Yrs    =P12_009N,
         Male22to24Yrs=P12_010N,
         Male25to29Yrs=P12_011N,
```

```
         Male30to34Yrs=P12_012N,
         Male35to39Yrs=P12_013N,
         Male40to44Yrs=P12_014N,
         Male45to49Yrs=P12_015N,
         Male50to54Yrs=P12_016N,
         Male55to59Yrs=P12_017N,
         Male60to61Yrs=P12_018N,
         Male62to64Yrs=P12_019N,
         Male65to66Yrs=P12_020N,
         Male67to69Yrs=P12_021N,
         Male70to74Yrs=P12_022N,
         Male75to79Yrs=P12_023N,
         Male80to84Yrs=P12_024N,
         MaleGE85Yrs  =P12_025N,
         TotFemale      =P12_026N,
         FemaleLT5Yrs   =P12_027N,
         Female5to9Yrs  =P12_028N,
         Female10to14Yrs=P12_029N,
         Female15to17Yrs=P12_030N,
         Female18to19Yrs=P12_031N,
         Female20Yrs    =P12_032N,
         Female21Yrs    =P12_033N,
         Female22to24Yrs=P12_034N,
         Female25to29Yrs=P12_035N,
         Female30to34Yrs=P12_036N,
         Female35to39Yrs=P12_037N,
         Female40to44Yrs=P12_038N,
         Female45to49Yrs=P12_039N,
         Female50to54Yrs=P12_040N,
         Female55to59Yrs=P12_041N,
         Female60to61Yrs=P12_042N,
         Female62to64Yrs=P12_043N,
         Female65to66Yrs=P12_044N,
         Female67to69Yrs=P12_045N,
         Female70to74Yrs=P12_046N,
         Female75to79Yrs=P12_047N,
         Female80to84Yrs=P12_048N,
         FemaleGE85Yrs  =P12_049N)

tract<-left_join(hh_tract,person_tract,
                 by=c("GEOID", "NAME"))%>%
  rename(Tract=GEOID)

# Similar code was used to prepare the blockgroup-level data.
```

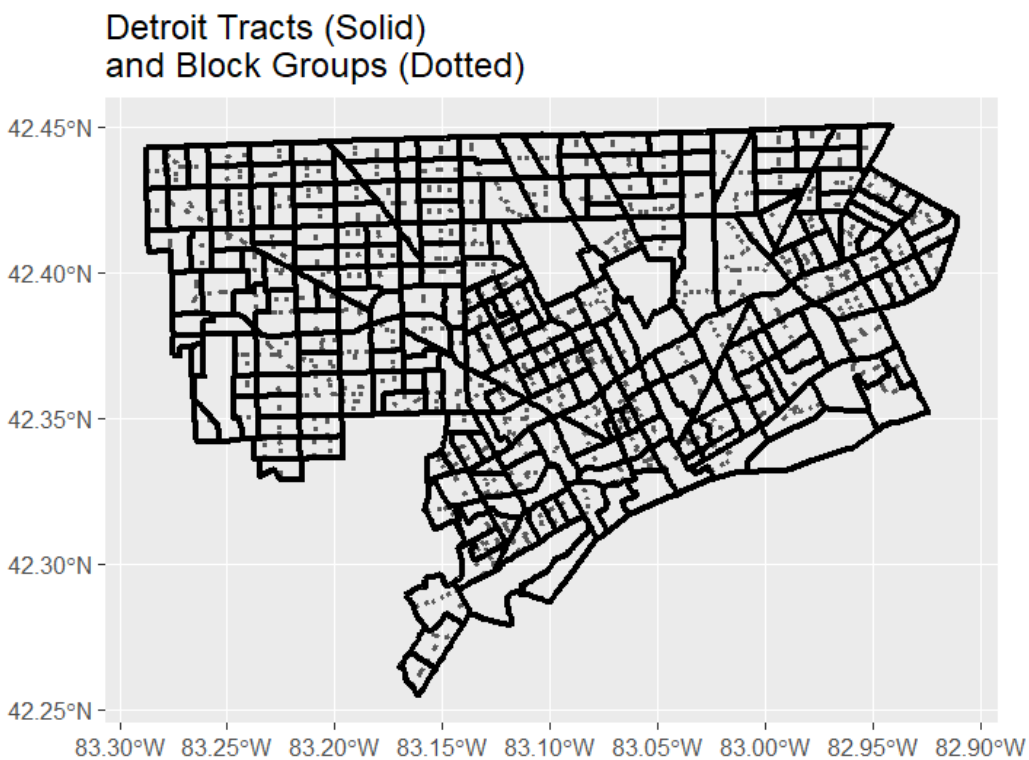
Maps of Detroit, MI



- Blue: Census tracts; Black: Census blockgroups
- The map is interactive, generated using the code below. It requires various packages. The main ones here are `tigris` and `leaflet`.

```
library(glue)
library(tigris)
library(sf)
options(tigris_class = "sf")
options(tigris_use_cache = T)
library(leaflet)

# To get PG tract map shapefile from tigris
Detroit_tract <- tracts(state="MI", county="Wayne", year=2020)%>%
  right_join(TractFIPS,by="GEOID")

st_crs(Detroit_tract) #Check the GIS projection type

Detroit_tract_trans <- st_transform(Detroit_tract,
                              "+proj=longlat +datum=WGS84")
st_crs(Detroit_tract_trans)
```

```r
# To get block group map shapefile
Detroit_bg <- block_groups(state="MI", county="Wayne", year=2020)%>%
  mutate(GEOIDt=substr(GEOID,1,nchar(GEOID)-1))%>%
  right_join(TractFIPS,by=c("GEOIDt"="GEOID"))%>%
  select(-GEOIDt)

Detroit_bg_trans <- st_transform(Detroit_bg, "+proj=longlat +datum=WGS84")

st_crs(Detroit_bg_trans)

# To create D blockgroup map layered with tracts
leaflet() %>%
  addTiles() %>%
  addPolygons(data=Detroit_tract_trans,
              popup = ~NAME,
              weight = 2,
              fillOpacity = 0.2,
              fillColor= "blue") %>%
  addPolygons(data=Detroit_bg_trans,
              popup = ~GEOID,
              color = "black",
              weight = 1,
              opacity= 1,
              smoothFactor = 1,
              fillOpacity = 0,
              fillColor= "snow")
```

- Simpler version

```
require(ggplot2)
ggplot() +
  geom_sf(data = Detroit_bg_trans, fill=NA,
          linetype="dotted", size=1) +
  geom_sf(data = Detroit_tract_trans, fill = NA,
          color = "black", size=1.2)+
  ggtitle("Detroit Tracts (Solid)
          \nand Block Groups (Dotted)")
```